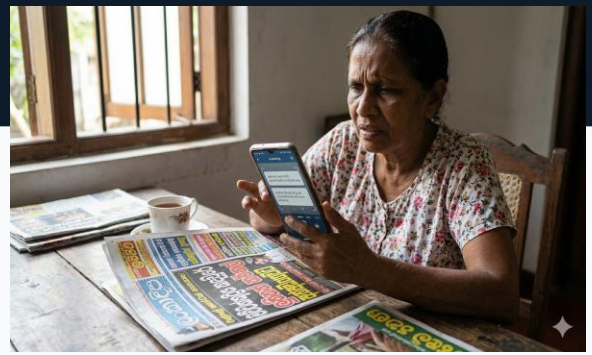
Interactive Sinhala Text Reader for the Visually Impaired with the assistance of OCR, NLP, and AI.

Group R26-DS-002 | Sri Lankan Institute of Information Technology

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Student ID	Comments	How these are addressed
IT22072238	• Need to improve communication. • Need to compare with existing work to validate. • What is the basis for proposing MAT Skeleton? • Noice correction from tortuosity.	• Existing work skip skeletonization and try to use the features direclty, but this may confuse the model when it comes to fonted text. • MAT is independent to the width and curvature of the letter's external walls • Noise correction done as blurring/smoothing and skeleton pruning.
IT22004468	• Need to compare with existing work to validate. • What the data requirement is.	• **STT:** Compared Whisper models, Wav2Vec2-BERT, Google STT selected based on WER measurement & **Intent Detection**: Compared (*NLLB+Llama3.2 vs direct Qwen2.5*) and selected NLLB approach based on accuracy and latency differences. • No static dataset required. The system dynamically builds a user history dataset in a local NoSQL database during runtime.
IT22259134	• Need to compare with existing work to validate. • Creating the 120k noisy-clean sentence pairs? • SinBERT for embedding or prediction? • mT5 for correction? • How do the models embed & work.	• Existing Sinhala correction systems are rule-based, word-level, and not OCR-specific; our work is the first context-aware transformer based Sinhala OCR post-correction system. • Stage 1 - SinBERT-Large for token-level OCR error prediction, • Stage 2 - mT5-small for contextual correction of only the flagged tokens. • Built a 120K noisy-clean Sinhala dataset (*30% real OCR ,70% synthetic noise*), 138 photographed newspaper pages & 1,883 manually annotated OCR errors.
IT22120670	• Need to improve communication. • Need to compare with exisitng work to validate. • Plan to retrieve the correct article if OCR fails • Classify a paragraph into levels based on difficulty to read • sinlin and LaBSE for embedding, Gemma 2B for generation -> why these models specifically?	• Existing system follow : **OCR -> text -> TTS** pipeline ,they output speech but **do not adapt content** to user comprehension needs. • OCR fail : System use 3 tier falls back to sufficient words to embed and query ChromaDB for the best-match article. • **Not** a full linguistic model **SVM classifier** uses 7 surface features to assign 3 difficulty levels, which **chooses generation mode** (simplify/elaborate). • **Sinling** – Sinhala Tokenization, **LaBSE** –Embedding model (sentence-transformer) • Instead of Gemma 2B, I selected Gemini 2.5 Flash via the Gemini API**.** because the task requires higher generation quality and more reliable instruction following.

The Problem and Solution



Problem

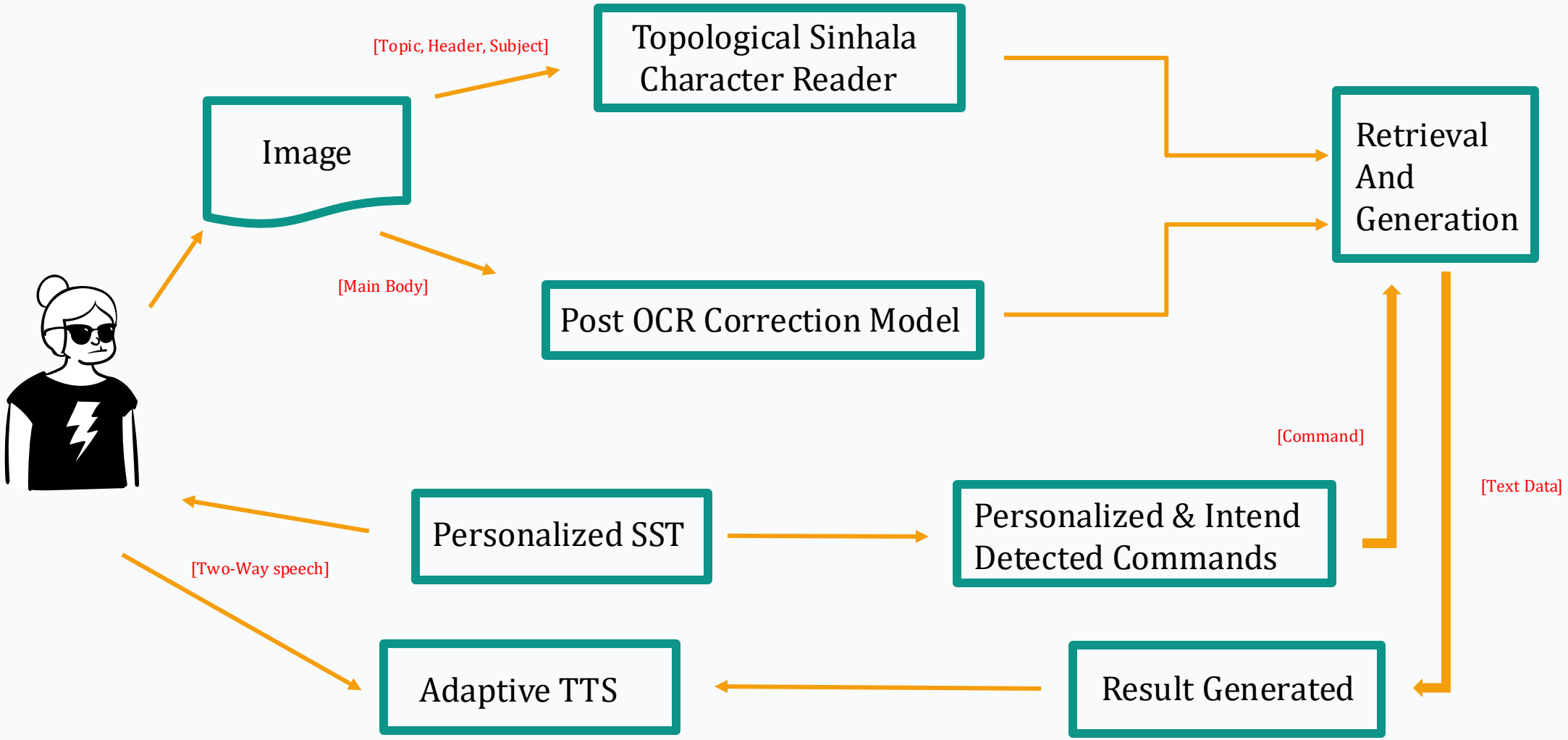
- People with Visual Impairment in Sri Lanka have limited access to information.
- Existing assistive tools like JAWS and Google Lookout are English-centric and offer no functional Sinhala script support.

Users and Stakeholders

- Primary users - visually impaired and elderly Sinhala-speaking individuals who rely on newspapers, textbooks and other documents for daily information.
- Secondary users - OCR developers and research institutions seeking a font-invariant Sinhala recognition, context aware Error correction, conversational Sinhala voice modules for integration.
- Tertiary stakeholders include Sri Lankan disability advocacy groups, government education agencies, and the open-source NLP community.

Visually impaired Sinhala speakers are excluded from printed information due to the absence of affordable, script-aware assistive technology.

Simplified System Overview & Flow



Individual Students' Components

M.J.J Muthunayaka (IT22072238)

Topological Sinhala Character Recognition via Medial Axis Transform

- Applies MAT-based skeletonization to reduce Sinhala characters to font-invariant structural representations for recognition.
- Segments and classifies headline and topical characters

H.P.I Madhusanka (IT22259134)

Context-Aware Sinhala Post-OCR Error Correction Model

- Detects OCR-corrupted tokens using a fine-tuned SinBERT binary classifier operating over full sentence context.
- Corrects flagged tokens using a fine-tuned mT5 sequence-to-sequence model to produce reliable Sinhala text output.

W.G.N Devindi (IT22120670)

Advanced Document Retrieval and Information Generation

- Retrieves relevant content from local databases, web sources, and previous scans using vector embeddings and semantic search.
- Generates simplified or elaborated responses via a RAG-powered Main LLM with cross-session context retention.

IT22004468

Adaptive Conversational Personalization and Voice Interaction

- Transcribes user speech, detects intent via a Mini LLM, and applies a personalization layer from stored user preferences.
- Delivers the final tailored response as natural Sinhala voice output via Text-to-Speech synthesis.

IT22072238 | MUTHUNAYAKA M.J.J



Topological Sinhala Character Recognition

via Medial Axis Transform

Topological Sinhala Character Recognition via Medial Axis Transform

The Problem

- Existing OCR tools fail on stylized **Sinhala headline fonts** where the **most information-rich** text in documents.
- No published system **specifically targets font-invariant** Sinhala display typography, beside from the few systems that just include them in the training data.



The Solution

- An OCR system designed to work on a **wide-variety of sinhala fonts** with font-invariant **skeletonization (MAT)** approaches for the training data as well as the input pipeline.

Prediction from a standard OCR	Truth Label
42% 2 රල 9 ල් ලර රල, 00! 02] ය,	අපගේ වංස කතාවල මෙන්ම සෙල් ලිපි වල, බොහෝ ගණනක් නිස්ස,
| දුවයේ, අන්ණම වන බොලවෙලා වග, බංතරිෂට අර වේ, ට	ඒ දවසේ අන්තිම ටික බොදවෙලා වගේ බංහරියට අර එකේ කොටු
—	මීතොටමුල්ලට ලක්ෂ බැගින් වන්දි මරික්කාර්ගෙන් සැර කතාවකුත් _
11[:1] [5[:11[566) ['1111511] (101	කලා ලොවේ උණුසුම් පුවත්
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Why Medial Axis Transform (MAT)

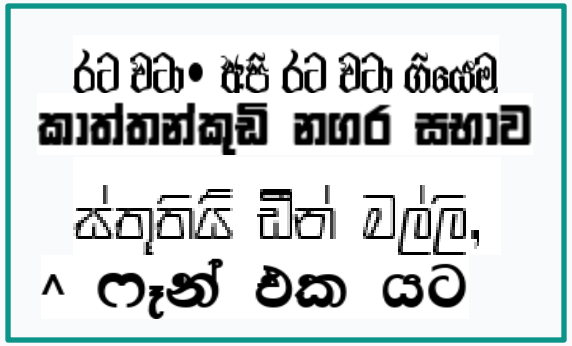
- Preserves topology being **robust** to **stroke variations.**
- **Lossless** compared to other skeletonization processes as it uses a **radius function** instead of using a destructive process
- Noise **Immunity** and Consistency compared to other thinning processes which are sensitive to **pixel noise** on character edges

Design and Process



[Pipeline 1]

[Joining of Pipeline 1 & 2 by using the below format]



[Pipeline 2]

Extracted text from the images

- Identify text boxes and crop
- Remove body text *(Implemented in a stricter manner)*
- Binarize, filter and skew *(finetune)*
- Split Horizontal by lines and append
- Resize to model input size

- Model Used : Efficient Net v2 with AdamW Optimization and CosineAnnealingLR.
- Class total : **309** with combination of **Regular** Letters, **Hal**-Version, **Combinations** as well **Pili** classes and few **punctuation** classes with **295** skeletons of different fonts per letter.
- Segmentation : True gaps, Slanted gaps, Valley projections, CCA blob detection - > whichever increase count is taken, but all methods were tested and choose.*(Account much harder for touching letters)*
- Recognition: Custom logic to include side-diatrics for better accuracy *(Check diatric logic for missed loopholes)*

IT22259134 Madhusanka H.P.I



Context-Aware Sinhala Post OCR Error Correction Model

for Assistive Document Reading

Context-Aware Sinhala Post OCR Error Correction Model

Research Gap - Existing vs This Work

System	Context-Aware	OCR-Specific	Sinhala
SinSpell 2021	✗	✗	✓
RoundTripOCR	✓	✗	✗
Dyslexia Pipeline 2025	✓	✗	✓
★ This Research	✓	✓	✓

Research Contributions

1st Formal 5-category Sinhala OCR error taxonomy (1883 real errors, 138 pages)

1st Domain-specific Sinhala OCR noise dataset (97,564 training pairs)

1st Transformer-based context-aware Sinhala post-OCR correction system

Two-Stage Pipeline Architecture

OCR Input *from Comp. 1* → **NFC Norm.** *Unicode clean* → **SinBERT** *ERROR / CORRECT* → **Threshold** *Confidence 0.65* → **mT5-small** *Seq2seq correct* → **Output** *Text + JSON conf*

Key Design: SinBERT gates mT5 — only [ERROR]-marked tokens are corrected. Clean tokens are NEVER touched → eliminates over-correction in TTS read-aloud systems.



Context-Aware Sinhala Post OCR Error Correction Model

Dataset

★ *First domain-specific Sinhala OCR noise dataset*

30%
Real OCR Pairs
138 newspaper pages
1883 manually annotated errors

70%
Synthetic Pairs
5-category noise engine
Replicates OCR glyph errors

After Cleaning: 97,564 train | 535 val *(ZWJ/ZWNJ removed — caused loss=0.0 training collapse)*

5 Category Sinhala OCR Error Taxonomy

Error Category	Sinhala Example	Freq.
Vowel Modifier Omission	ගෙදර → ගදර	46.9%
Hal-sign Misdetection	කෙටි → කෙටී	18.3%
Visual Substitution	ශ → ෂ	14.7%
Compound Glyph Split	ක්ෂ → ක+ෂ	11.2%
Font Encoding Noise	→ random	8.9%

Evaluation Results

System	CER	WER	Over-Corr.
B1 — Raw OCR	0.0367	0.1048	—
B2 — Rule-based	0.0367	0.1048	—
B3 — mT5 only	0.0104	0.0249	0.0%
★ **Proposed**	**0.0319**	**0.0862**	**1.3%**

Proposed vs B1: CER ↓13.1% | WER ↓17.8% | Over-correction 1.3%

Threshold Sensitivity Analysis

Threshold	CER	vs B1	Note
0.65 (val-tuned)	0.0212	+3.2%	*Primary — reported in paper*

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Advanced Document Retrieval and Information Generation

RAG Pipeline for Sinhala Assistive Reading — Component 3

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Advanced Document Retrieval and Information Generation

The Problem

- Existing Sinhala assistive reading system follow a narrow **OCR -> text -> TTS** pipeline ,they output speech but **do not adapt content** to user comprehension needs.

- This component adjusts the text dynamically by simplifying dense content for some users while elaborating simplified content for others - ensuring comprehension across all users

The Solution

- **Sinhala Content Adaptation Pipeline** combining an SVM-based text classifier with a tiered retrieval confidence pipeline and Gemini-powered RAG generation to dynamically **simplify** or **elaborate** content in Sinhala,
- Ensuring that a single scanned document can be meaningfully delivered to users across the full spectrum of content-comprehension needs, from users requiring simplified explanations to domain experts expecting precise, information-dense output.

User Requirements Address by The Solution

- Users receive **contextually relevant information** matched to the scanned document content, not generic or off-topic results.
- The system delivers **comprehension-level-adapted output** simplified for low-complexity users, elaborated for advanced users

Design Process

Stage 1 – Input Handler

- **Parses Component 2 JSON:** sentence level tokens, corrected text, error flags
- **Filters error tokens & builds 3 text tiers :**

  Tier 1 – reliable only

  Tier 2 – corrected

  Tier 3 – raw OCR
- **Improve retrieval , reduce OCR noise**

Stage 2 & 3 – Retrieval + Chroma DB

- **Embedding model:** LaBSE / sentence-transformer
- **ChromaDB articles collection** : returns best-match article + top-k candidates + similarity score
- **Tier selection:** prefers Tier 1 → falls back to Tier 2 → Tier 3 if needed *(Check similarity threshold for retrieval confidence)*

Stage 4 – SVM Classifier

- **7 features:** avg sentence length, avg word length, type–token ratio, formal connective density, formal verb ending ratio, punctuation ratio, long-word ratio
- **SVM** → assigns **Level 1 / 2 / 3**
- Combines level + **Component 4 intent signal** to decide final generation

Stage 5 - RAG Generation (Gemini API)

- Builds **prompt** using : retrieved article + predict level + user params from component 4.
- Calls **Gemini 2.5 Flash via the Gemini API** in RAG setup → grounded on selected article
- Produces Sinhala response adapted to difficulty.

Output JSON → Component 4

- Adapted Sinhala text
- Assigned difficulty level · retrieval confidence · predict level
- Diagnostics: OCR error rate, tier used, candidate article list

IT22004468 | SATHSARA S.P.Y B



Adaptive Conversational Personalization and Voice Interaction Module

(Personalization and Voice Interaction Module)

Component 4 — Interactive Sinhala Text Reader for the Visually Impaired

Adaptive Conversational Personalization and Voice Interaction Module

How the Solution Addresses the Sub-Problem / Prototype

01 **The Gap:** Current Sinhala assistive readers rely on static, one-way Text-to-Speech (TTS), forcing users to passively listen without the ability to ask questions or adjust complexity.

02 **Dynamic Adaptation:** The system adjusting the final Sinhala TTS output - reading speed, detail level, and vocabulary - based on the user's request.

User Requirements Addressed by the Solution

01 **Natural Conversational Control:** Allows visually impaired users to navigate by using their natural Sinhala voice.

02 **Contextual Flexibility:** Satisfies the requirement for adaptive reading (e.g., simplifying academic text for an elderly user or reading quickly for a student).

Voice Input & Preprocessing

- Capture Input:
- STT: Google's Internal STT
- Translation: Sinhala to English via local **NLLB-600M**



Intent & Personalization

- Local LLM (Llama 3.2 1B via Ollama)
- Intent Extraction
- Behavioral Profiling: Fetches user's profile from **database**
- Query Construction : intent + personalization flags into a prompt for the Main LLM



Adaptive Voice Output

- Format Response
- Adaptive Sinhala TTS: adjusting the speech rate and tone to match the user's profile

Adaptive Conversational Personalization and Voice Interaction Module

Design Excellence / Contribution

- **Personalization engine** a local NoSQL database to log runtime interactions. A background LLM process analyzes the user history and dynamically update the user's profile.
- **NLP Architecture:** lack of Sinhala NLU datasets introduced a dual step pipeline (SinhalaSTT ➡ NLLB Translation ➡ LLM). This reduces token consumption and latency on standard hardware.

User Feedback on Prototype

- **Conversational Freedom:** The user appreciated they can say "explain this simply" their own words.

Future Implementations

- Implement the background LLM analyzing and automatically update user preference profiles in the local database.
- Finalize the JSON output for the Main RAG LLM.
- Develop the Sinhala TTS layer to dynamically adjust speech rate and complexity based on the user profile.

Customer Persona

- The **primary** intended **users** -> **visually impaired people** who would like to adaptively interact with Sinhala text documents.
- If **expanded** into **Mainstream** OCR, -> much larger user base of all Sinhala OCR system users.

Market Size

- **Primary** market size is small, with reports of 1.7% having Visual Impairment (https://rda.sliit.lk/handle/123456789/2788).
- **Secondary** user market can be large, as certain component could be implemented to other mainstream systems

Uniqueness of solution

- First of this kind of device specifically dedicated to Visually impaired Sinhala readers (most are English)
- The technology tackles issues that may normally be explored, but not much in the Sinhala language(fonts, errors, Intents, retrievals)

Recoverability & Sustainability

- The topological character reader can be implemented to update with latest fonts
- OCR induced errors rarely change unless improved on by externally, in which the model's workload will decrease, making it faster & more accurate.
- The System dynamically learns user preferences from runtime interactions.

Thank You!

Group R26-DS-002 | Sri Lankan Institute of Information Technology

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